

1. Brainstorming & Ideation Phase

- **App Concept:** Create an end-to-end AI platform that translates simple user prompts or sketches into fully structured, multi-panel comic books.
- **Feature Scope:** Outline core features like automated scriptwriting, multi-agent story generation, genre and art style selection, dialogue generation, and visual character consistency.

2. Requirement Analysis Phase

- **Google Generative AI Stack:**
 - **Gemini Models:** Generates 3-act scripts, panel-by-panel descriptions, camera angles, and character dialogue.
 - **Imagen:** Handles high-quality panel art generation based on detailed visual prompts generated by Gemini.
- **Backend & Infrastructure:** Plan REST/gRPC API integration using Google Cloud Functions or FastAPI, with Google AI Studio for prompt prototyping.

3. Project Design Phase

- **UI/UX Mockups:** Design user input controls for genre selection (e.g., Sci-Fi, Fantasy, Noir), character profiles, and panel layouts (e.g., 2-panel strip, 4-panel grid).
- **Architecture Flowchart:** Define how story prompts move from user input → Gemini Script Generation → Visual Prompting & Imagen Rendering → Canvas Compositing.

4. Project Planning Phase

- **Milestones & Tasks:** Divide tasks across sprint cycles (e.g., setup Gemini API, prompt engineering, canvas UI integration, speech bubble overlay).
- **Resource Allocation:** Set up GitHub version control, environment variable security for API keys, and model latency benchmarks.

5. Project Development Phase

- **Story & Script Engine:** Integrate Gemini API to construct character arcs, punchy dialogue, sound effect text, and precise panel directions.
- **Visual Rendering Pipeline:** Use Imagen to convert scene descriptions into comic illustrations maintaining style coherence.
- **Layout & Speech Bubble Compositing:** Build dynamic canvas overlays to render speech bubbles, thought bubbles, captions, and panel gutters programmatically.

6. Project Testing Phase

- **Character & Style Consistency:** Evaluate prompt techniques to ensure character design stays recognizable across consecutive panels.
- **Performance & Edge Cases:** Test API response times, dialogue fitting within speech bubbles, and user edit flows.

7. Project Documentation Phase

- **Developer Docs:** Write a comprehensive README.md detailing API configurations, model setup, and environment variables.
- **User & API Guides:** Document prompt structures, character creation guidelines, and exported file options (PDF, PNG, Webtoon format).

8. Project Demonstration Phase

- **Deployment:** Deploy the web application on Google Cloud Run or showcase it through Google AI Studio.
- **Live Demo:** Create a video walkthrough or interactive demo showing a raw idea transforming into a finished, readable comic.